MATH1014 Cheat Sheet

November 11, 2026

1 Divergence/Convergence tests

D	Diverge
C	Converge
I	Inconclusive

p-series

$$p > 1 \Rightarrow \sum \tfrac{1}{n^p} = C, \quad p \leq 1 \Rightarrow D$$

Integral

$f > 0$ continuous decreasing, $a_n = f(n)$

$$\int_1^\infty f(x)\,\mathrm{d}x = C/D \Leftrightarrow \sum a_n = C/D$$

Comparison

$$0 \leq a_n \leq b_n, \sum b_n = C \Rightarrow \sum a_n = C$$

$$0 \leq b_n \leq a_n, \sum b_n = D \Rightarrow \sum a_n = D$$

Limit comparison

$$a_n, b_n > 0, \quad \lim_{n\to\infty} \frac{a_n}{b_n} = c, \quad 0 < c < \infty$$

$$\sum b_n = C \Rightarrow \sum a_n = C$$

$$\sum b_n = D \Rightarrow \sum a_n = D$$

Alternating series

$$\sum (-1)^{n-1} b_n, \quad b_n > 0$$

b_n decreasing, $b_n \to 0 \Rightarrow C$

Absolute convergence

$$\sum |a_n| = C \Rightarrow \sum a_n = C$$

Ratio

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

$$L < 1 \Rightarrow C, \quad L > 1 \Rightarrow D, \quad L = 1 \Rightarrow I$$

Root

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$$

$$L < 1 \Rightarrow C, \quad L > 1 \Rightarrow D, \quad L = 1 \Rightarrow I$$

2 Lagrange Multipliers

Equation

$$\nabla f = \lambda \nabla g$$

Example

$g(x,y) = x^2 + y^2 = 2$

$$\begin{pmatrix}2\\2\end{pmatrix} = \lambda \begin{pmatrix}2x\\2y\end{pmatrix}$$

$$\left(\frac{1}{\lambda}\right)^2 + \left(\frac{1}{\lambda}\right)^2 = 2$$

$$\begin{pmatrix}2\\2\end{pmatrix} = 1 \times \begin{pmatrix}2x\\2y\end{pmatrix}$$

$f(x,y) = 2x + 2y$

$$x = y = \frac{1}{\lambda}$$

$$\lambda = \pm 1$$

$$\begin{pmatrix}2\\2\end{pmatrix} = -1 \times \begin{pmatrix}2x\\2y\end{pmatrix}$$

3 Jacobian

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

Change of variables

$$\iint_R f(x,y)\,\mathrm{d}A$$

$$= \iint_S f(x(u,v), y(u,v))\ |J|\,\mathrm{d}u\,\mathrm{d}v$$

4 Least Squares

Normal Formula

$$A^T A\hat{x} = A^T b$$

Example

From the points $\begin{pmatrix}0\\1\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}2\\2\end{pmatrix}$

$$A = \begin{pmatrix}1 & 0\\1 & 1\\1 & 2\end{pmatrix} \qquad b = \begin{pmatrix}1\\2\\2\end{pmatrix}$$

5 Partial Differentiation

Critical Point

$$f_{x(x,y)} = 0, f_{y(x,y)} = 0$$

Second Derivatives Test

$$D = f_{xx} f_{yy} - (f_{xy})^2$$

Interpretation of D

$D > 0, f_{xx} > 0$	local min
$D > 0, f_{xx} < 0$	local max
$D < 0$	saddle
$D = 0$	inconclusive

6 Polar Integrals

Equations

$$A = \int_\alpha^\beta \frac{1}{2} r^2\,\mathrm{d}\theta$$

$$L = \int_\alpha^\beta \sqrt{r^2 + \left(\mathrm{d}\frac{r}{\mathrm{d}\theta}\right)^2}\,\mathrm{d}\theta$$

Double Integral

$$\iint_D f(x,y)\,\mathrm{d}A$$

$$= \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta) r\,\mathrm{d}r\,\mathrm{d}\theta$$

7 Plane/Line Equations

Line

$P_0 = (x_0, y_0, z_0)$ $\qquad \vec{v} = (a,b,c)$

Vector	$\boldsymbol{r} = \boldsymbol{r}_0 + t\boldsymbol{v}$
Parametric	$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$
Symmetric	$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

Plane

$P_0 = (x_0, y_0, z_0)$ $\qquad \vec{n} = (a,b,c)$

Vector	$\boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0) = 0$
Scalar	$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$
Linear	$ax + by + cz = d, \quad d = ax_0 + by_0 + cz_0$

Sphere

$\vec{c} = (h,k,l)$ $\qquad R =$ radius

Vector	$\lvert\boldsymbol{r} - \boldsymbol{c}\rvert = R$
Dot product	$(\boldsymbol{r} - \boldsymbol{c}) \cdot (\boldsymbol{r} - \boldsymbol{c}) = R^2$
Standard	$(x-h)^2 + (y-k)^2 + (z-l)^2 = R^2$
Expanded	$x^2 + y^2 + z^2 + Ax + By + Cz + E = 0$